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Availability: Aug. 2026 - Jul. 2027
San Francisco Bay Area

WORK AUTHORISATION

Singapore Citizen, eligible for **12-month J-1 US visa**, visa-sponsored by NUS, hosted by Stanford University.

EDUCATION

National University of Singapore | Bachelor of Computing in **Computer Science** 2027

NUS Scholar | Residential College Entrepreneurship Club **President** | Notion **Campus Leader** | **Teaching Assistant**

EXPERIENCE

ByteDance Singapore

Applied ML Orchestration Intern (ML Infrastructure) | k8s, Ray, PyTorch, Go, Scheduling Apr. 2026 - Aug. 2026

- Building ML infrastructure for **fault-tolerant autoscaling** and resource optimisation across **heterogeneous GPU/CPU clusters** on Exabyte-scale training data for TikTok's global Search, Ads, & Recommendation models.
- Implementing **backpressure-driven** VPA control across distributed training frameworks using throughput, GPU utilisation, & task lag to detect regressions & trigger automated rollback, preventing SLA violations.
- Designing stage-aware **autoscaling logic** to detect ML training lifecycle stages (warmup, failover, restart) and trigger lifecycle-aware, **in-place scale-down** strategies via a **custom CRD-based controller**.
- Developing guaranteed in-place VPA scale-up for critical pods by integrating with the Kubernetes **eviction layer** to preemptively drain lower-priority workloads, leveraging **QoS-class-aware** eviction policies and admission webhooks.

Software Engineering Intern (BytePlus Platform Engineering) | Go, Node.js, RAG, Next.JS Jan. 2026 - Mar. 2026

- Deployed features for byteplus.com's Seedream5.0 launch, driving conversions of **1k+ B2B clients** & **440k+ MAU**.
- Led 0→1 development of BytePlus **RAG** search: deployed **Go-driven Firecrawlers** for **ETL data ingestion** of **5,000+ documentation pages**, integrated vector retrieval with SSE multimodal rendering for LLM responses.

IBM (Client Engineering) Singapore

Full Stack Developer Intern | Spring Boot, Node.js, AWS, SQL Server, CrowdStrike, React Sep. 2025 - Dec. 2025

- **Forward deployed** on-site to co-engineer **B2G** SG military personnel **RBAC** system with 65k+ DAU.
- Built **Lambda**-automated event & visitor retry management batch pipeline using **Node.js** workers, integrating **FALCON** & Sequelize for clearance, triggering notifications using **AWS SNS/SES/SQS**.
- Implemented end-to-end conditional & automated on/offboarding with real-time **cross-system validation** to handle user **role management** across multiple locations, reducing manual data entry by **80%**.

CloudJoi (Malaysia's #1 arts ticketing platform) - cloudjoi.com Kuala Lumpur, Malaysia

Software Engineering Intern | Laravel, React Native, Next.js May. 2025 - Aug. 2025

- Contributed towards AI **semantic** search & recommendations via **vector embedding** & clustering metadata.
- Launched knowledgebase.cloudjoi.com—dashboard helpdesk for **200+ clients**—0→1 on **Next.js** with RESTful API.
- Integrated **password change confirmation** flow with **queue-based email notifications** & Redis caching.

PROJECTS

sgl-project OSS Contributions | *Python, PyTorch, SGLang, HuggingFace, Speculative Decoding* May 2026 - Present

- Generalised spec-V1 hidden_states slicing across EAGLE/FROZEN_KV_MTP in SGLang #26217, **fixing silent token undercount** via per-request cumsum offsets
- Implemented **end-to-end standalone speculative decoding** (Leviathan/Chen 2023) in mini-sglang #134: `verify_drafts()` with parametrised tests, byte-equality validated vs HF **assisted generation** on Qwen3-1.7B/0.6B.
- Engineered **opt-in FP8 quantised (float8_e4m3fn) KV cache pool** in mini-sglang #132 doubling cache capacity (verified **2.00x** on Qwen3) with **21.1% throughput** & **26.8% median ITL gains** at conc=20 on RTX 4060.

Grid Universe AI Agent | *PyTorch, NumPy, Scikit-learn* Oct. 2025 - Nov. 2025

- Engineered an autonomous agent for a stochastic grid-world, across 48 complex test levels.
- Designed a 2-layer CNN architecture (PyTorch) to reconstruct game state from snapshots with **99% accuracy**.

TECHNICAL SKILLS

ML Systems: PyTorch, SGLang, HuggingFace transformers, FlashAttention, Quantisation, KV cache, speculative decode

Distributed & Infra: Kubernetes, Ray, Docker, heterogeneous GPU clusters, autoscaling, training orchestration

Languages: Python, Go, C++, C, TypeScript, Java, SQL, Bash

Cloud & Other: AWS, GCP, Git, Linux, NumPy, pandas, Next.js, React, Spring Boot, FastAPI, Matplotlib